

Lucas Leonel Galardo Euler

San Telmo, Buenos Aires, Argentina · Remote

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Professional Summary

Gameplay and systems programmer in **Unity (C#)** with **10+ years** of development experience: gameplay mechanics, modular AI (FSM, steering, GOAP, pathfinding), motion and game feel, and reusable systems (custom player loop, GPU instancing with Jobs/Burst, editor tooling). When a project calls for it, I go **low-level** (C++, DirectX 11, HLSL/compute shaders). I maintain my own library of ~20 reusable Unity packages that I use to prototype and ship fast. **Top of class (9.69/10)** in Virtual Simulator Design & Programming — Da Vinci.

Professional Experience

Technical Consultant, Unity (*Part-time, partner track*) — Somos Leyenda

Remote · Spain | 2025 – Present - **Audit** third-party Unity codebases (architecture, scripts, networking, code quality, dependencies, scenes) and deliver canonical documentation plus prioritized **P0–P3** improvement plans. - **Detect** structural bugs and race conditions in existing pipelines (static state not reset across scenes, sync errors); implement critical fixes and refactors when the audit flags them as priority. - **Co-organized** the Somos Leyenda × Lifecole Game Jam (Mar 2026); the winner was one of my own students.

Unity Instructor / Developer (*Part-time*) — Lifecole

Remote · Madrid, Spain | 2021 – Present - **Authored** the *Unity Proficiency* curriculum (the advanced Unity track) and lead the advanced courses: OOP in C#, design patterns, Burst/Jobs optimization and HLSL shaders. - **Built** an automated, ScriptableObject-based unit-testing system for standardized student evaluation (2026). - **Develop** internal Unity tools and interactive experiences for education.

Freelance Game Developer / Consultant

Remote | 2015 – Present - **Prototype** 2D/3D games in Unity (URP, Shader Graph, HLSL): mechanics, tools and polish. - **Built** multiplayer with Photon Fusion 2 and custom production systems for clients (mobile in React Native, bots, SQL-backed payment/accounting systems). - **Mentored** 40+ private students across games, web and algorithms.

Tech Lead / Product Engineer (*Part-time*) — Nexio

Remote · Spain | 2026 – Present - **Lead end-to-end** a multi-channel SaaS platform ([nexio.land](#)): serverless architecture, full-stack implementation, CI/CD (feature → staging → production), roadmap and sprint prioritization. (*Current technical-leadership anchor; core stack here is web/TypeScript, not games.*)

Software Developer (*Full-time*) — Wilson Champ S.R.L.

CABA, Argentina | 2015 – 2021 - **Developed** embedded software in C and Assembler (Freescale microcontrollers) with low-level optimization; IoT with Arduino and ESP8266. - **Built** full-stack web in **PHP/MySQL over several years** (JavaScript/jQuery/AJAX, HTML5/CSS3) on Apache, plus database administration and domain/DNS/cPanel infrastructure.

Selected Projects

La Maldición: Héroes de Lorthar — competitive card game (Unity, in development)

Official digital adaptation of an Argentine card game (physical version sold in **40+ stores**, “best friendship-breaker” award 2022). Publicly playable WebGL build: itch.io. Built a **manual binary serialization subsystem over Netcode for GameObjects** ($O(1)$ indexing by PlayerID, zero allocations on the hot path) that replaced JSON in the multiplayer transport.

Common-Package — reusable Unity systems library (2021 – Present)

~20 in-house packages: custom **player loop** over Unity’s native `PlayerLoop`, **GPU instancing** with Jobs+Burst, hierarchical **FSM** with runtime Lowest-Common-Ancestor resolution, a **binary heap with Decrease-Key** that brings Dijkstra/A* down to $O(E \log V)$, lite DI via code generation, and editor tooling.

Unity HLSL compute-shader compilation on DirectX 11 — Image Campus (2025)

Native **C++ plugin** that compiles HLSL compute shaders at runtime on DX11 (x86+x64), with $C++ \leftrightarrow C\#$ bindings and an automatic fragment→compute converter. Diagnosed a silent GPU black-screen bug down to **3 root causes**: UAV recreation per frame, a main→render thread race, and 16-byte constant-buffer misalignment.

Arrange the Heaven — Da Vinci thesis (2025)

Procedural RPG in Unity (team; role: programmer & designer): procedural systems, modular AI (FSM/steering/GOAP) and custom URP rendering. **Mass toggle of 4,096 GameObjects** with a hybrid Jobs + main-thread architecture (parallel state computation via Jobs; `SetActive` on the main thread for API thread-safety).

Technical Skills

Languages: C# · C++ · C · HLSL · GLSL · Lua · Assembler · JavaScript · TypeScript · SQL · PHP

Engine & graphics: Unity (URP, Shader Graph, Burst, Jobs, ScriptableObjects, editor tooling) · DirectX 11 + native plugins · compute shaders · Netcode for GameObjects · Photon Fusion 2

Game programming: modular AI (FSM, steering, GOAP) · pathfinding (Dijkstra/A/Theta) · GC-free data structures · performance optimization (Jobs/Burst) · procedural generation · unified motion (linear + angular)

Web (full-stack): PHP · MySQL · JavaScript · HTML5/CSS3 · Apache · React · TypeScript · Node.js · Supabase

Embedded / IoT: C & Assembler (Freescale) · Arduino · ESP8266

Tooling & methods: Git · CI/CD · editor tooling · automated unit testing · code review · technical documentation

Education

Virtual Simulator Design & Programming — Da Vinci, Argentina (2021–2025) · *Top of class, GPA 9.69/10.*

Shaders for Video Games — Image Campus (2025) · *HLSL/GLSL; capstone: C++ compute-shader library on DX11.*

Electronics Technician — Colegio San José A-355 (2008–2014).

Languages

- **Spanish:** Native
- **English:** Professional Working Proficiency (advanced technical reading, professional writing, technical meetings)